Lesson 2.3 - Cumulative Relative Frequency Table

> **Video:** How To Make a Cumulative Relative Frequency Table by The Organic Chemistry Tutor
> **Chapter:** 2 - Graphs and Tables

Idea

Cumulative relative frequency (CRF) is a **running total** of the relative frequencies. For each value, it gives the fraction of the data that is **less than or equal to** that value.

$$\mathrm{CRF}_k = \mathrm{RF}_1 + \mathrm{RF}_2 + \cdots + \mathrm{RF}_k$$

Steps

1. Columns: **V** (value), **f** (frequency), **RF** (relative frequency), **CRF** (cumulative relative frequency).
2. Fill in V and f, then find $n = \sum f$.
3. RF = f/n. Check that the RF column adds to 1.
4. CRF: start with the first RF, then keep adding the next RF.
5. **Check:** the last CRF must be **1.00**.

Example

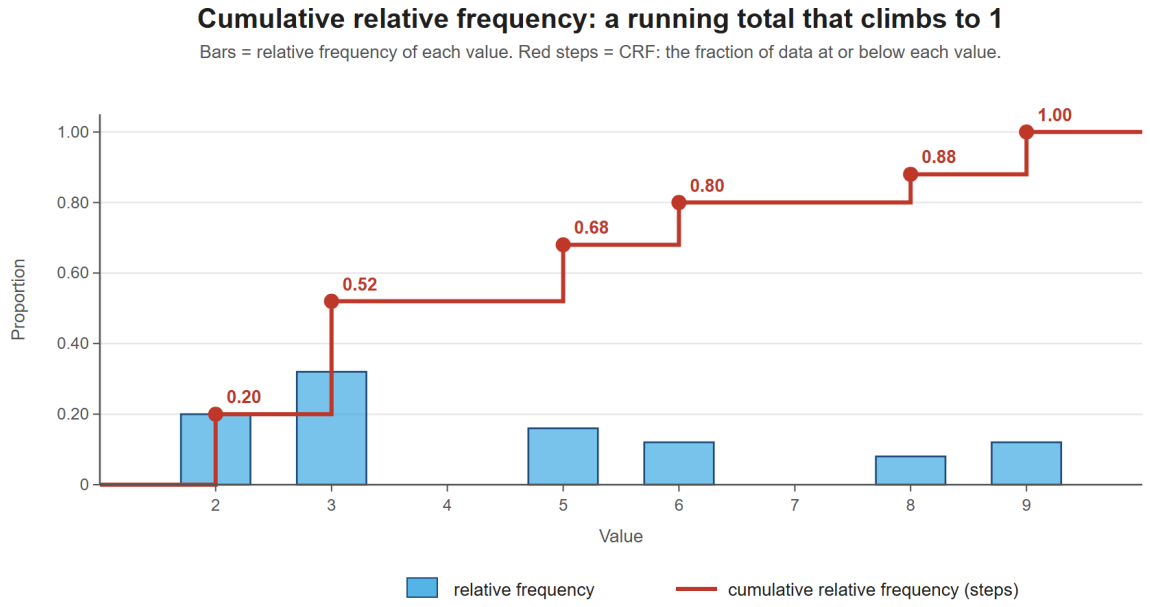
The data has 25 values with these counts (there are no 4s or 7s):

V	f	RF = f / 25	CRF
2	5	0.20	0.20
3	8	0.32	0.20 + 0.32 = **0.52**
5	4	0.16	0.52 + 0.16 = **0.68**
6	3	0.12	0.68 + 0.12 = **0.80**
8	2	0.08	0.80 + 0.08 = **0.88**
9	3	0.12	0.88 + 0.12 = **1.00**
Total	**25**	**1.00**	

Reading the CRF column:

- 0.52 at V = 3 → 52% of the values are 3 or less.

- 0.80 at V = 6 → 80% of the values are 6 or less.



The cumulative relative frequency only changes at data values, so it is drawn as steps: flat until the next value, then a jump equal to that value's relative frequency. It never goes down and it ends at 1.00. Read it as "the fraction of the data at or below this value" (e.g. 0.80 of the values are 6 or less).

Key takeaways

- CRF = running sum of RF. It never decreases and always ends at **1**.
- CRF answers "what fraction of the data is **at or below** this value?"
- That makes the CRF column useful for **percentiles and quartiles** (see Lesson 1.1 and Chapter 4).